

DA5000W Linear algebra and optimization

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Tutorial 5 Optimization

1 Unconstrained Optimization

1.1 Theoretical Recap

Unconstrained optimization refers to the problem

$$\min_{x \in \mathbb{R}^n} f(x),$$

where the objective function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is differentiable (and often assumed twice differentiable).

First-Order Necessary Condition: If x^* is a local minimizer and f is differentiable at x^* , then

$$\nabla f(x^*) = 0.$$

Second-Order Sufficient Condition: If f is twice differentiable, then:

$$\nabla^2 f(x^*) \succ 0$$

1.2 Example on Unconstrained Minimization

Consider the function

$$f(x, y) = 2x^2 + 3y^2 - 4x + 6y + 5, \quad (x, y) \in \mathbb{R}^2.$$

We will find stationary points using the first-order condition and classify it using the second-order condition.

1.2.1 Step 1: Gradient (first-order necessary condition)

Compute the gradient:

$$\nabla f(x, y) = \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \end{pmatrix} = \begin{pmatrix} 4x - 4 \\ 6y + 6 \end{pmatrix}.$$

Set $\nabla f(x, y) = 0$:

$$4x - 4 = 0, \quad 6y + 6 = 0.$$

Solve:

$$x^* = 1, \quad y^* = -1.$$

So $(x^*, y^*) = (1, -1)$ is the only stationary point.

The function value at the optimal point is zero in this case.

1.2.2 Step 2: Hessian (second-order test)

Compute the Hessian matrix:

$$H(x, y) = \nabla^2 f(x, y) = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 6 \end{pmatrix}.$$

Note: The eigenvalues are 4 and 6 (both positive), so H is positive definite.

1.2.3 Conclusion

By the first-order condition we found the stationary point $(1, -1)$. Since the Hessian at that point is positive definite, the second-order sufficient condition guarantees that $(1, -1)$ is a *strict local minimizer*. Because the function is a strictly convex quadratic, this minimizer is global.

1.2.4 Short additional examples

1. $g(x, y) = -2x^2 - y^2$. Gradient $\nabla g = (-4x, -2y)$ so stationary point at $(0, 0)$. Hessian $H = \begin{pmatrix} -4 & 0 \\ 0 & -2 \end{pmatrix}$ is negative definite $\Rightarrow (0, 0)$ is a strict local *maximum*.
2. $h(x, y) = x^2 - y^2$. Gradient zero at $(0, 0)$. Hessian $H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$ is indefinite $\Rightarrow (0, 0)$ is a *saddle point*.

1.3 Gradient Descent

The Gradient descent algorithm uses the following update rule (with $\alpha > 0$) to minimise a given function.

$$x_{k+1} = x_k - \alpha \nabla f(x_k).$$

1.4 Example on Gradient Descent

Consider the convex quadratic function

$$f(x) = (x - 3)^2.$$

It has gradient

$$f'(x) = 2(x - 3),$$

and unique minimizer $x^* = 3$.

1.4.1 Gradient descent update

Given a stepsize (learning rate) $\alpha > 0$,

$$x_{k+1} = x_k - \alpha f'(x_k) = x_k - \alpha 2(x_k - 3).$$

We choose $\alpha = 0.25$ and initial point $x_0 = 0$.

Thus,

$$x_{k+1} = x_k - 0.5(x_k - 3) = 0.5x_k + 1.5.$$

1.4.2 Compute the iterates

$$x_1 = 0.5(0) + 1.5 = 1.5,$$

$$x_2 = 0.5(1.5) + 1.5 = 2.25,$$

$$x_3 = 0.5(2.25) + 1.5 = 2.625,$$

$$x_4 = 0.5(2.625) + 1.5 = 2.8125,$$

$$x_5 = 0.5(2.8125) + 1.5 = 2.90625.$$

We see the iterates move halfway from the current point to the minimizer 3 each time.

1.4.3 Table of iterates

Iteration k	x_k	$f(x_k) = (x_k - 3)^2$
0	0	9.0000
1	1.5	2.2500
2	2.25	0.5625
3	2.625	0.1406
4	2.8125	0.0352
5	2.90625	0.0088

1.5 Newton's Method Recap

For a twice differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$, Newton's method iteratively updates

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}.$$

It is derived by using second order Taylor approximation.

1.6 Example: Newton's Method

Consider the quadratic function

$$f(x) = \frac{1}{2}x^\top Qx + c^\top x, \quad Q = \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}, \quad c = \begin{pmatrix} -2 \\ -4 \end{pmatrix}, \quad x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

1.6.1 Step 1: Gradient and Hessian

Since $f(x) = \frac{1}{2}x^\top Qx + c^\top x$ with $Q = Q^\top$, we have

$$\nabla f(x) = Qx + c, \quad \nabla^2 f(x) = Q.$$

Explicitly,

$$\nabla f(x) = \begin{pmatrix} 2x_1 - 2 \\ 8x_2 - 4 \end{pmatrix}, \quad \nabla^2 f(x) = \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}.$$

1.6.2 Step 2: Newton direction

The Newton direction d_k satisfies

$$Qd_k = -\nabla f(x_k).$$

Since $\nabla f(x_k) = Qx_k + c$, this becomes

$$Qd_k = -(Qx_k + c) \quad \implies \quad d_k = -x_k - Q^{-1}c.$$

1.6.3 Step 3: Compute $Q^{-1}c$

The inverse of Q is

$$Q^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{8} \end{pmatrix}.$$

Thus

$$Q^{-1}c = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{8} \end{pmatrix} \begin{pmatrix} -2 \\ -4 \end{pmatrix} = \begin{pmatrix} -1 \\ -0.5 \end{pmatrix},$$

so that

$$-Q^{-1}c = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}.$$

1.6.4 Step 4: Newton update

With full Newton step $\alpha_k = 1$,

$$x_{k+1} = x_k + d_k = x_k - x_k - Q^{-1}c = -Q^{-1}c = \begin{pmatrix} 1 \\ 0.5 \end{pmatrix}.$$

1.6.5 Conclusion

- The iterates satisfy $x_{k+1} = (1, 0.5)^\top$ for *any* starting point x_0 .
- Thus Newton's method reaches the minimizer in **one iteration**.
- This occurs because f is a quadratic function and its second-order Taylor expansion is exact.
- Since $Q \succ 0$, the minimizer $x^* = (1, 0.5)$ is unique.

2 Constrained Optimization

2.1 Equality and Inequality Constraints

General constrained optimization problems have the form

$$\min_x f(x) \quad \text{s.t.} \quad g_i(x) \leq 0, \quad i = 1, \dots, m, \quad h_j(x) = 0, \quad j = 1, \dots, p.$$

2.2 Lagrangian

The *Lagrangian* associated with the problem is

$$L(x, \lambda, \mu) = f(x) + \sum_{i=1}^m \lambda_i g_i(x) + \sum_{j=1}^p \mu_j h_j(x),$$

where

$$\lambda_i \geq 0 \quad (\text{multipliers for inequality constraints}), \quad \mu_j \in \mathbb{R} \quad (\text{multipliers for equality constraints}).$$

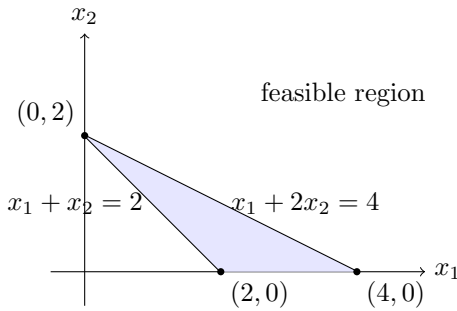
2.3 KKT Conditions

If f , g_i , and h_j are differentiable, the KKT conditions state that at an optimal point x^* , there exist multipliers $\lambda_i \geq 0$ and μ_j such that

$$\begin{aligned} \nabla f(x^*) + \sum_{i=1}^m \lambda_i \nabla g_i(x^*) + \sum_{j=1}^p \mu_j \nabla h_j(x^*) &= 0, \\ g_i(x^*) &\leq 0, \\ h_j(x^*) &= 0, \\ \lambda_i g_i(x^*) &= 0. \end{aligned}$$

2.4 Example: Constrained Optimization

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & f(x) = x_1 + 2x_2 \\ \text{s.t.} \quad & g_1(x) := -x_1 - x_2 + 2 \leq 0, \\ & g_2(x) := x_1 + 2x_2 - 4 \leq 0, \\ & g_3(x) := -x_1 \leq 0 \quad (\Leftrightarrow x_1 \geq 0), \\ & g_4(x) := -x_2 \leq 0 \quad (\Leftrightarrow x_2 \geq 0). \end{aligned}$$



(Geometrically, the feasible set is the triangle with vertices $(2, 0)$, $(4, 0)$, $(0, 2)$.)

2.4.1 Lagrangian

Introduce multipliers $\lambda_i \geq 0$ for g_i . The Lagrangian is

$$L(x, \lambda) = x_1 + 2x_2 + \lambda_1(-x_1 - x_2 + 2) + \lambda_2(x_1 + 2x_2 - 4) + \lambda_3(-x_1) + \lambda_4(-x_2).$$

2.4.2 KKT conditions

1. Stationarity: $\nabla_x L(x, \lambda) = 0$:

$$\begin{aligned} \frac{\partial L}{\partial x_1} : 1 - \lambda_1 + \lambda_2 - \lambda_3 &= 0, \\ \frac{\partial L}{\partial x_2} : 2 - \lambda_1 + 2\lambda_2 - \lambda_4 &= 0. \end{aligned} \tag{S}$$

2. Primal feasibility: $g_i(x) \leq 0$ for all i .
3. Dual feasibility: $\lambda_i \geq 0$ for all i .
4. Complementary slackness: $\lambda_i g_i(x) = 0$ for each i .

2.4.3 Solve the KKT system

We first identify the active constraints at the (vertex) candidates. Check the vertex $x^* = (2, 0)$:

$$\begin{aligned} g_1(2, 0) &= -2 - 0 + 2 = 0 \quad (\text{active}), & g_2(2, 0) &= 2 + 0 - 4 = -2 \quad (\text{inactive}), \\ g_3(2, 0) &= -2 < 0 \quad (\text{inactive}), & g_4(2, 0) &= 0 \quad (\text{active}). \end{aligned}$$

By complementary slackness, the multipliers for inactive constraints must be zero:

$$\lambda_2 = 0, \quad \lambda_3 = 0.$$

Substitute into stationarity (S):

$$\begin{aligned} 1 - \lambda_1 + 0 - 0 &= 0 \quad \Rightarrow \quad \lambda_1 = 1, \\ 2 - \lambda_1 + 0 - \lambda_4 &= 0 \quad \Rightarrow \quad 2 - 1 - \lambda_4 = 0 \quad \Rightarrow \quad \lambda_4 = 1. \end{aligned}$$

These multipliers satisfy dual feasibility ($\lambda_1 = \lambda_4 = 1 \geq 0$) and complementary slackness:

$$\lambda_1 g_1(2, 0) = 1 \cdot 0 = 0, \quad \lambda_4 g_4(2, 0) = 1 \cdot 0 = 0.$$

Primal feasibility is already checked.

2.4.4 Conclusion

All KKT conditions hold for

$$\boxed{x^* = (2, 0), \quad \lambda^* = (\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (1, 0, 0, 1)}.$$

The optimal objective value is

$$f(x^*) = x_1^* + 2x_2^* = 2.$$

2.4.5 Remarks

- Since this is a linear program and the KKT conditions are necessary and sufficient (convex problem, linear constraints), the KKT solution above is a global optimum.
- Geometrically the objective is minimized at the vertex $(2, 0)$ where the constraint $x_1 + x_2 \geq 2$ and the nonnegativity $x_2 \geq 0$ are active.

2.5 Example: QP

Consider the convex problem

$$\begin{aligned} \min_{x \in \mathbb{R}^2} \quad & f(x) = x_1^2 + x_2^2, \\ \text{s.t.} \quad & h(x) := x_1 + x_2 - 1 = 0 \quad (\text{equality}), \\ & g(x) := 0.8 - x_1 \leq 0 \quad (\text{inequality, i.e. } x_1 \geq 0.8). \end{aligned}$$

This is convex (f is strictly convex, constraints linear), so KKT conditions are *necessary and sufficient* for optimality.

2.5.1 Lagrangian

Introduce multiplier $\lambda \in \mathbb{R}$ for the equality and $\mu \geq 0$ for the inequality. The Lagrangian is

$$\mathcal{L}(x, \lambda, \mu) = x_1^2 + x_2^2 + \lambda(0.8 - x_1) + \mu(x_1 + x_2 - 1).$$

2.5.2 KKT conditions

1. Stationarity: $\nabla_x \mathcal{L} = 0$:

$$\frac{\partial \mathcal{L}}{\partial x_1} : 2x_1 + \mu - \lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial x_2} : 2x_2 + \mu = 0.$$

2. Primal feasibility: $x_1 + x_2 - 1 = 0$, $0.8 - x_1 \leq 0$.

3. Dual feasibility: $\lambda \geq 0$.

4. Complementary slackness: $\lambda(0.8 - x_1) = 0$.

2.5.3 Solve the KKT system

We check whether the inequality is active at the optimum. There are two possibilities:

Case A: inequality inactive $\Rightarrow \lambda = 0$. Then stationarity gives

$$2x_1 + \mu = 0, \quad 2x_2 + \mu = 0 \quad \Rightarrow \quad x_1 = x_2.$$

Using the equality $x_1 + x_2 = 1$ we get $x_1 = x_2 = \frac{1}{2}$. But $x_1 = \frac{1}{2} < 0.8$, so the inequality $x_1 \geq 0.8$ is violated. Hence Case A is infeasible.

Case B: inequality active $\Rightarrow 0.8 - x_1 = 0$ and $\lambda \geq 0$. From the equality constraint we then have

$$x_1 = 0.8, \quad x_2 = 1 - x_1 = 0.2.$$

Substitute into stationarity:

$$2(0.8) + \mu - \lambda = 0 \quad \Rightarrow \quad 1.6 + \mu - \lambda = 0,$$

$$2(0.2) + \mu = 0 \quad \Rightarrow \quad 0.4 + \mu = 0 \quad \Rightarrow \quad \mu = -0.4.$$

From the first equation,

$$1.6 + (-0.4) - \lambda = 0 \quad \Rightarrow \quad \lambda = 1.2 > 0.$$

Complementary slackness holds (since $0.8 - x_1 = 0$ and $\lambda > 0$). All KKT conditions are satisfied.

2.5.4 Conclusion

The unique optimal solution is

$$x^* = (0.8, 0.2)^\top, \quad \lambda^* = 1.2, \quad \mu^* = -0.4$$

and the optimal objective value is

$$f(x^*) = 0.8^2 + 0.2^2 = 0.64 + 0.04 = 0.68.$$

2.6 Equality-Constrained Quadratic Program

Consider the equality-constrained QP

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad & f(x) = \frac{1}{2} x^\top Q x + c^\top x, \\ \text{s.t.} \quad & Ax = b, \end{aligned}$$

where $Q = Q^\top \succeq 0$.

2.6.1 Lagrangian

The Lagrangian of the problem is

$$\mathcal{L}(x, \lambda) = \frac{1}{2}x^\top Qx + c^\top x + \lambda^\top (Ax - b).$$

2.6.2 KKT Conditions

The KKT optimality conditions are:

1. **Stationarity:**

$$Qx + c + A^\top \lambda = 0.$$

2. **Primal feasibility:**

$$Ax = b.$$

2.6.3 Newton System for Solving KKT Conditions

Newton's method applied to the KKT system gives the linear system

$$\begin{pmatrix} Q & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} \Delta x \\ \Delta \lambda \end{pmatrix} = - \begin{pmatrix} Qx + c + A^\top \lambda \\ Ax - b \end{pmatrix}.$$

2.6.4 Newton Update

After solving for $(\Delta x, \Delta \lambda)$, the update is

$$x_{k+1} = x_k + \Delta x, \quad \lambda_{k+1} = \lambda_k + \Delta \lambda.$$

2.6.5 One-Step Convergence

Because the KKT system of the quadratic program is *linear*, Newton's method with full step size $\alpha_k = 1$ solves the system in **one iteration**.

Reference Material

Refer to the following book for problems on Optimization-

<https://www.math.uci.edu/~qnie/Publications/NumericalOptimization.pdf>